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Analysis of Generalized Semiparametric Regression Models for Cumulative Incidence Functions with Missing Covariates

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1 Preliminary - Competing Risks Data

- Study participants are exposed to two or more mutually exclusive causes of failure.

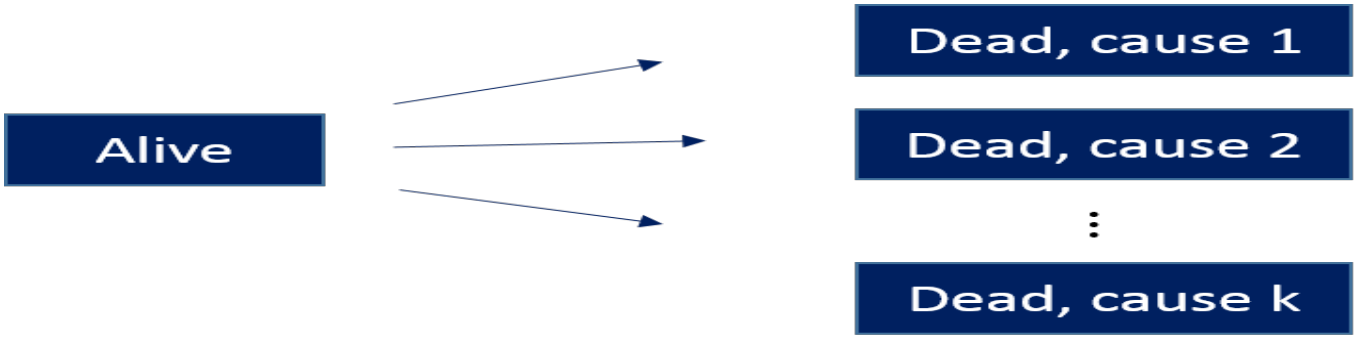


Figure 1: Each subject may die from k different causes.

- The failure time T_i with k different failure types $J_i \in \{1, 2, \ldots, k\}$ under the right censoring time C_i, $i = 1, \ldots, n$.

Continue on Competing Risks Data

- The observed right-censored competing risks data (iid):

$$Y_i = (X_i, Z_i, \tilde{T}_i, \tilde{J}_i),$$

  where $\tilde{T}_i = \min(T_i, C_i)$, $\Delta_i = I(T_i \leqslant C_i)$ and $\tilde{J}_i = J_i \Delta_i$.

- $\tilde{J}_i = 1$: the failure time is observed at $\tilde{T}_i$ and the cause of failure 1.

2 A Preventive HIV-1 Vaccine Efficacy Trial (RV 144)

- 16,394 HIV-1 negative volunteers were randomized to

  the vaccine group ($n = 8198$) and placebo group ($n = 8196$).

- Subjects were vaccinated at weeks 0,4,12 and 24.

- Monitored them for 42 months for the occurrence of HIV-1 infection after immune response biomarkers were measured from vaccine recipients at Week 26 visit.

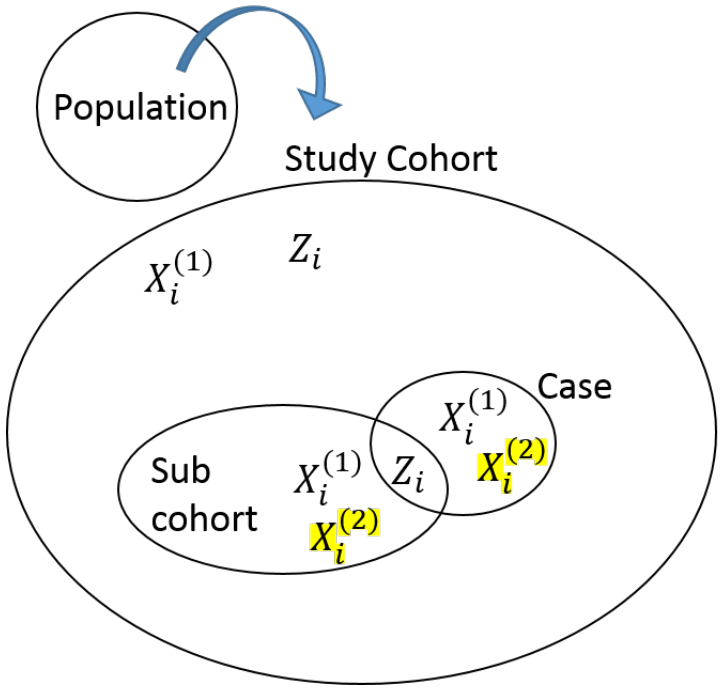
- Examine the effects of immune response biomarkers on the cumulative probability of becoming infected with specific generic type of HIV-1.

Issue on Measuring Biomarkers

- Immune response biomarkers are not observable for all subjects, even from failures events.
- Rate of HIV infection is very low (less than 1%).

  $\Rightarrow$ no need to measure the biomarkers from all vaccine recipients.
- Measured by generalized Case-Cohort design (Prentice, 1986).

Case-Cohort Design in the Form of Two-Phase Sampling



- Given covariates (X_i, Z_i),

  X_i has two parts $(X_i^{(1)},\ X_i^{(2)})$.

- The phase two covariate data: $X_i^{(2)}$

  (e.g., immune response biomarkers).

- The phase one covariate data: $X_i^{(1)}, Z_i$

  (e.g., observed infection time, treatment type, behavioral risk scores, gender, and age).

- Under Missing At Random (MAR) assumption (Rubin, 1976),

  $\Rightarrow$ Two Phase Sampling leads to missingness of covariate X_i.

Main Objective

- The conditional cumulative incidence function $F_1(t; X_i, Z_i) = P(T_i \leqslant t, J_i = 1 | X_i, Z_i)$ is modeled by the generalized semiparametric model (Scheike et al. (2008)):

$$F_1(t; X_i, Z_i) = h\{X_i^T \eta(t) + g(\gamma, Z_i, t)\} \ , i = 1, \ldots, n$$

where $h(\cdot)$ is a known link function and $g(\cdot)$ is a known function.

  - Estimate $\boldsymbol{\eta}(t)$ and $\boldsymbol{\gamma}$ when some covariates of X_i are missing.
    - $\eta(t)$: time dependent regression coefficients.
    - γ: time-invariant coefficients.
  - Proposed estimating procedure using an inverse probability weighting (IPW) of complete cases estimating equations for $\boldsymbol{\eta}(t)$ and $\boldsymbol{\gamma}$.

3 IPW estimating equations under Missing Data

- The IPW estimating equations

  $= \xi_i/\varphi(\mathcal{V}_i, \theta) \times$ Full data weighted least square estimating equations,

  where $\varphi(\mathcal{V}_i, \theta)$ is a sampling probability and ξ_i is a selection indicator of the phase-two sample.

- Idea of Inverse Probability Weighted Method

Subject (i =1,2,3,4)	1	2	3	4
Missingness of X_i	Missing	Missing	Missing	Selecting
IPW $\xi_i/\varphi(V_i, \theta)$	0	0	0	1/0.25 = 4

- The IPW estimate $\widehat{\boldsymbol{\eta}}(t)$ and $\widehat{\gamma}$ are consistent estimators if the selection probability is correctly specified.

4 Apply IPW Procedure to Vaccine Group in RV144

Estimate the cumulative incidence of HIV-1 Infection with cause $k = 1, 2$

$$F_k(t; X_i, Z_i) = 1 - \exp(-\{\eta_{k0}(t) + \eta_{k1}(t)R_i + \gamma_{k1}B_{1i}t + \gamma_{k2}B_{2i}t\}),$$

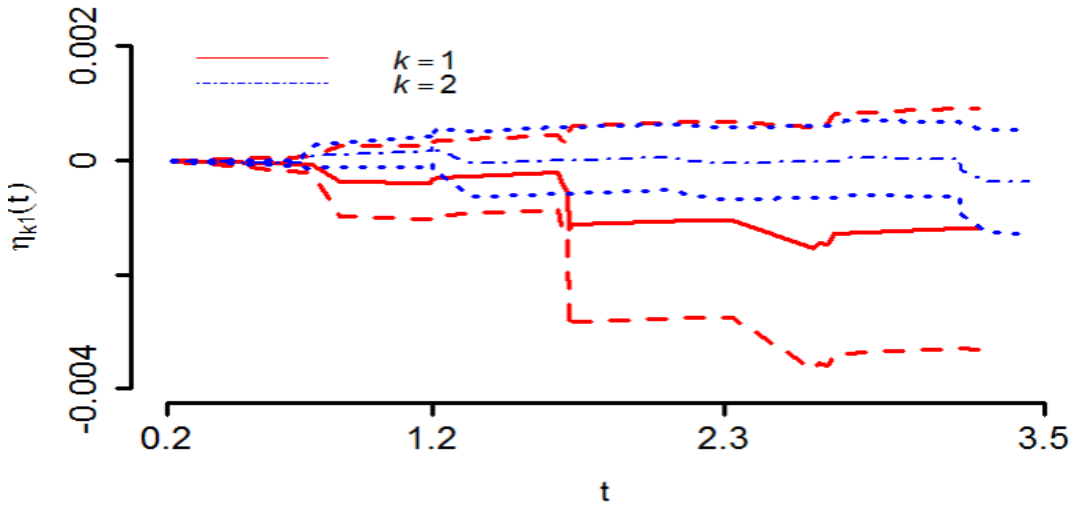
where $X_i = (1, R_i)^T$, $Z_i = (B_{1i}, B_{2i})^T$.

- Phase-two covariate: the immune response R_i (IgG-A244V1V2)
- In the two-phase sampling, R_i was measured in 83% (34 of 41) of cases and 2.5% (205 of 8157) of controls.
- Competing HIV-1 genetic types of interest :

  Shorter Mark $(\tilde{J}_i = 1)$ vs. Larger Marks $(\tilde{J}_i = 2)$

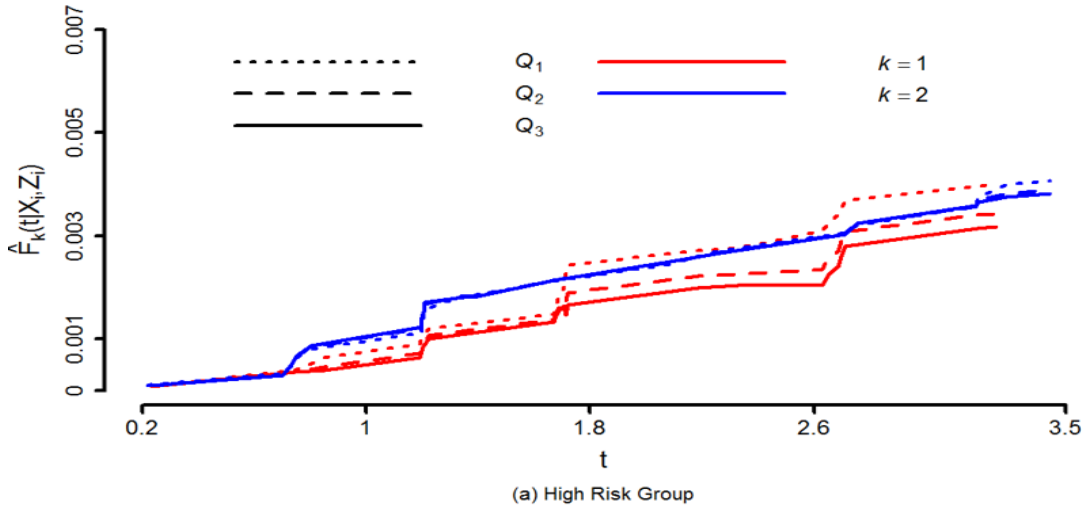
  where mark variable which is the generic distance of infecting HIV sequences to the HIV sequences injected by the vaccine.

Estimated Effects of Immune Responses (IgG-A244V1V2)



- Negatively correlated with the cumulative incidence of HIV infection with shorter marks: A244 would be important for protection of HIV infection.

Cumulative Probability of HIV-1 Infection with Competing Causes



- $Q_1 = -0.1, Q_2 = 0.3, Q_3 = 0.5$ are quartiles of immune responses R_i.
- A higher immune responses may be associated with a lower risk of HIV-1 infection with near A244 vaccine insert. $\Rightarrow$ The A244 induced antibodies are a cause of vaccine efficacy.

5 Conclusions

- The IPW methods are modifications of the full data weighted least squares equations by weighting to the observed members through the inverses of estimated sampling probabilities.
- The information can not be fully utilized because of excluding the subjects having missing observations.
- This research provides insights into the potential beneficial effects of IgG binding antibody response for different types of infecting HIV-1 viruses.
- Further investigation is needed to improve efficiency and robustness.

6 Acknowledgments

Co-authors

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- Thomas H. Scheike – Department of Biostatistics, University of Copenhagen
- Peter B. Gilbert – University of Washington, Vaccine and Infectious Disease and Public Health Sciences Divisions at Fred Hutchinson Cancer Research Center

Thank you!

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